The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 2 MODULE, AUTUMN SEMESTER 2020 - 2021

OPERATING SYSTEMS AND CONCURRENCY
COMP 2035 (G52OSC)

Time allowed TWO Hours

An additional 30 minutes to cover IT issues and upload time

Answer TWO questions out of THREE

Submit your answers containing all the work you wish to have marked as a **single PDF file, with each page in the correct orientation, to the appropriate dropbox on the module's Moodle page.**

Use the standard naming convention for your document: [Student ID]_[Module Code]. Write your student ID number at the top of each page of your answers. Do not include your name.

Although you may use any notes or resources you wish to help you complete this open-book examination, the academic misconduct policy that applies to your coursework also applies here. You must be careful to avoid plagiarism, collusion or false authorship. This statement refers to, and does not replace, the University policy which stipulates severe penalties for academic misconduct.

Staff are not permitted to answer assessment or teaching queries during the period in which your open-book examination is live. If you spot what you think may be an error on the exam paper, note this in your submission but answer the question as written.

IMPORTANT: Lateness penalties will not be applied, and late submission will not be accepted.

1. Process Scheduling and Memory Management (25 Marks)

(a) The Shortest Remaining Time First (SRTF) is a variant of Shortest Job First (SJF) with pre-emption added in. Consider the following five processes vying for the CPU. The scheduler uses SRTF. The scheduler re-evaluates which process to run only upon the arrival of a new process into the scheduling queue, or the completion of a process. The table shows the arrival time of each process.

Process	Arrival time	Execution time
P1	T0	6ms
P2	To + 2ms	5ms
P3	To + 3ms	1ms
P4	To + 4ms	4ms
P5	To + 5ms	2ms

The scheduling starts at time T0. Fill in the table below with the process that is executing on the processor during each time slot.

Time Interval T0+	0	1	2	3	.	.	.		
Process Running	.	.	.	.	.				

Time Interval T0+	.	.	.	.					
Process Running	.	.	.	.					

Use the tables with process and remaining time to show your work as to how you arrived at the above schedule.

[5 Marks]

(b) How would you prevent starvation in a multi-level queue scheduling algorithm?

[2 Marks]

(c) What scheduling policy would you normally choose for the following systems? Why?

(i) A time sharing system

(ii) A batch system [2 Marks]

(d) Briefly describe how you could implement the *LRU* page replacement algorithm in hardware. Is this algorithm the best in all cases? Why or why not?

[4 Marks]

(e) Consider the following page reference string:

1, 3, 1, 2, 3, 4, 2, 3, 1, 2, 3, 4

Assume there are three (3) physical page frames that are initially empty. Show how many page faults would occur for the Least Recently Used (LRU) page replacement algorithms. Show the details of your work. [6 Marks]

(f) During the time interval t1 - t2, the following virtual page accesses are recorded for the three processes P1 and P2, respectively.

P1: 0, 0, 1, 2, 1, 2, 1, 1, 0

P2: 0, 100, 101, 102, 103, 0, 1, 2, 3

(i) What is the working set for each of the above three processes for this time interval? [2 Marks]

(ii) What is the cumulative memory pressure on the system during this interval?

[1 Mark]

(g) Consider a swapping system for which the free list indicates the following blocks of memory (a) 10KB, (b) 15KB, (c) 12KB, (d) 21KB, (e) 30KB, (f) 9KB, (g) 16KB (in that order). Assume that requests for blocks of memory of 12Kb, 9Kb and 25Kb come in (in that order). Which partitions would be allocated for the first fit, next fit, and best fit algorithm?

[3 Marks]

2. Concurrency and Deadlock (25 Marks)

(a) Briefly describe what mutexes and semaphores are, and what their key differences are. [3 Marks]

(b) Are the wait and signal operations on semaphores implemented as atomic operations? Briefly explain and illustrate your answer (e.g. using pseudo code). [3 Marks]

(c) Consider the code below involving two threads P0 and P1.

```
#include <stdio.h>
#include <pthread.h>

int turn = 0;
int sum = 0;

void * P0(void * a)
{
        int i;
        for(i = 0; i < 100000; i++)
        {
                while(turn != 0);
                        sum++;
                turn = 0;
                // other code here
        }
}

void * P1(void * a)
{
        int i;
        for(i = 0; i < 100000; i++)
        {
                while(turn != 1);
                        sum++;
                turn = 1;
                // other code here
        }
}

int main()
{
        pthread_t tP1, tP0;
        pthread_create(&tP0, NULL, P0, NULL);
        pthread_create(&tP1, NULL, P1, NULL);
        pthread_join(tP0, NULL);
        pthread_join(tP1, NULL);
        printf("%d\n", sum);
}
```

(i) Determine whether the code is synchronised in a correct manner. State how you would change the code if this is not the case. [2 Marks]

(ii) Explain whether the code (or the modified code) meets all requirements for mutual-exclusion. [2 Marks]

(d) Critically evaluate the *strict alternation algorithm* with reference to the mutual exclusion problem.

[6 Marks]

(e) What reasonable safeguards might be built into an operating system to prevent processes from waiting indefinitely for an event?

[4 Marks]

(f) The fact that a state is *unsafe* does not necessarily imply that the system will be in *deadlock*. Explain why this is true.

[2 Marks]

(g) Consider the following snapshot of a system in the Table 2. There are no current outstanding queued unsatisfied requests.

Resource Usage

Processes	Current Allocation				Maximum Demand				Resources Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P1	0	0	1	2	0	0	1	2	2	1	0	0
P2	2	0	0	0	2	7	5	0				
P3	0	0	3	4	6	6	5	6				
P4	2	3	5	4	4	3	5	6				
P5	0	3	3	2	0	6	5	2				

Table 2.

Answer the following questions using banker's algorithm.

(i) Calculate the number of resources needed to satisfy the maximum demand of the processes p1, p2, p3, p4 and p5.

[1 Mark]

(ii) Is this system in a safe state? [1 Mark]

(iii) If a request from process P3 arrives for (0, 1, 0, 0), can that request be granted immediately?

[1 Mark]

3. Process Scheduling, Concurrency, Memory Management, File Management and Device Management (25 Marks)

(a) This question relate to device management:

Disk requests are received in the order 48, 21, 1, 24, 21, 39, 77, 6 and 60. The starting position for a disk arm is at track 12. Given a disk with 80 tracks (0:79) With the help of diagram, calculate the number of tracks crossed when the following algorithms are used:

(i) First-come, first served.

(ii) Shortest Seek (Time) First.

(iii) The elevator (SCAN) algorithm when the arm is moving down.

[6 Marks]

(b) This question relates to i-node file systems:

(i) Describe the implementation and principles behind a file system with i-nodes (give one example). [4 Marks]

(ii) Explain how does an i-node file system resolve the shortcomings of other file systems (e.g. contiguous, FAT)? [2 Marks]

(c) This question relates to input/output sytems.

Describe various types of buffering in the I/O subsystem of an operating system. Give reasons why it is required, and give a case where it is an advantage, and a case where it is a disadvantage.

[6 Marks]

(d) This question relates to concurrent programming.
Why is it considerably more difficult to test, debug, and prove program correctness for concurrent programs than for sequential programs?

[4 Marks]

(e) This question relates to real-time systems:

What are the conditions and restrictions that you would want to impose on a multitasking system that is being used with real-time process?

[3 Marks]